06

Adapter 调控 DeepGen DiT 的架构

每个 denoise step 通过 forward pre-hook 动态生成 6 组残差；不修改冻结的 DeepGen 源代码

蓝=输入 绿=编码 紫=推理
橙=Adapter 红=Detail 灰=冻结

A 每步输入与共享嵌入

ControlledDeepGenPipeline
transformer.register_forward_pre_hook

step k / K
progress = k / (K-1)
timestep [B]

target noisy latent [B,16,Hlat,Wlat]
DeepGen patch embed (patch=2)
→ target_hidden [B,Ntarget,1536]

text hidden [B,Ntext,4096]
context embedder → [B,Ntext,1536]
pooled text [B,2048] + timestep → temb

分支 context
Geometry: text + global adapter tokens
Interaction: text + global adapter tokens
Detail: text + detail tokens（独立）

三份空间 condition
Geometry 144ch • Interaction 144ch
Detail 144ch
各自独立 condition_embed → 1536

B SharedRecurrentControlCore：三分支 × 六级递归

Geometry Branch
全样本运行
独立 hidden/context state

Interaction Branch
index_select
双人有效行
单人输出保持 0

Detail Branch
仅 detail_valid 行
face / hand 共享

branch hidden0 = target_hidden + condition_embed(condition)
branch context0 = 对应 context

Stage s = 0…5
① + stage_embedding[s]
② 共享 DeepGen joint transformer block（复制自 block 0）
③ + 独立 LowRankStageAdapter 1536→64→1536
④ CrossNorm(control_hidden, target_hidden)
⑤ zero-init Linear head → residual group s
⑥ 更新本分支 hidden 与 context，进入下一 stage

递归 ×6

每 stage 复用

共享权重
pos_embed
time_text_embed
context_embedder
1× joint transformer block

分支独立参数
condition embed
6 stage embeddings
6 rank-64 adapters
6 CrossNorm + 6 zero heads

BranchControlResiduals
geometry / interaction / optional detail
各 6×[B,Ntarget,1536]

C 强度、对齐与 24-block DiT 注入

ControlStrengthController / group s
$R_s = G_s \cdot$geometry_strength$\cdot$schedule
$+ I_s \cdot$interaction_strength$\cdot$schedule
$+ D_s \cdot$region_gate$\cdot$detail_strength$\cdot$detail_schedule
所有 group gate = exp(log_scale) > 0

Detail region_gate
face = max(person face masks) × face_strength
hands = max(person,left/right masks) × hand_strength
region_gate = max(face, hands)
detail schedule: cosine, progress 0.35→1

align_target_residuals
把 [B,Ntarget,1536] 对齐到 DeepGen 完整 token 顺序
source-reference 与 padding token 位置补 0
→ 6×[B,Nfull,1536]

block_controlnet_hidden_states

Active block mapping
Group 0 → blocks 0–3
Group 1 → blocks 4–7
Group 2 → blocks 8–11
Group 3 → blocks 12–15
Group 4 → blocks 16–19
Group 5 → blocks 20–22
Block 23: context_pre_only，不注入

冻结 DeepGen 内部效果

每个被映射的 DiT block 在自己的 hidden stream 中加上对应 residual；Adapter 通过 hook 提供控制，不改 Transformer 权重。detail 缺失或 detail_strength=0 时跳过 Detail Branch，保持 V6.3 行为。